

AGDF-GenConViT: Adaptive Gated Decision Fusion for Deepfake Detection

Rayan Ahmed

Dept. of Artificial Intelligence
FAST School of Computing, NUCES
Islamabad, Pakistan
i230018@isb.nu.edu.pk

Awwab Ahmed

Dept. of Artificial Intelligence
FAST School of Computing, NUCES
Islamabad, Pakistan
i230079@isb.nu.edu.pk

Uwaid

Dept. of Artificial Intelligence
FAST School of Computing, NUCES
Islamabad, Pakistan
i232574@isb.nu.edu.pk

Rana Zohaib

Dept. of Artificial Intelligence
FAST School of Computing, NUCES
Islamabad, Pakistan
i230096@isb.nu.edu.pk

Abstract—Detection of deepfake videos is an increasingly critical problem in digital forensics as generative models continue to improve in producing convincing synthetic faces. The hybrid model GenConViT [1] employs two branches—an Encoder-Decoder (ED) variant and a Variational Autoencoder (VAE) variant—yet fuses their predictions via a static 50/50 average that disregards per-sample branch reliability. This paper presents AGDF-GenConViT, which replaces the static averaging with an *Adaptive Gated Decision Fusion* (AGDF) module inspired by the Adaptive Gated Fusion Network (AGFN) originally proposed for multimodal sentiment analysis [2]. The AGDF module is a lightweight gated MLP ($\approx 17.9K$ parameters) that receives the 4-dimensional concatenated probability vector $[P_{ed}, P_{vae}]$ from the two frozen backbone branches and produces dynamic per-sample fusion weights (w_{ed}, w_{vae}) via a Softmax output layer. Training is performed on the FaceForensics++ dataset using Kaggle Dual T4 GPUs, with an ablation study across Cross-Entropy and Focal Loss ($\gamma \in \{0.5, 1.0, 2.0, 3.0, 5.0\}$) and a Leave-One-Deepfake-Out (LODO) protocol for generalization evaluation. AGDF-GenConViT with Focal Loss ($\gamma=2.0$) achieves 97.1% accuracy, F1 = 0.968, and AUC-ROC = 0.991 on the test set, outperforming the static-average baseline by +1.7% accuracy and improving out-of-distribution accuracy from 72.3% to 79.4% under LODO evaluation.

Index Terms—deepfake detection, adaptive fusion, gated MLP, GenConViT, FaceForensics++, focal loss, out-of-distribution generalization

I. INTRODUCTION

The rapid advancement of AI-driven image generation has produced increasingly realistic facial manipulation techniques—deepfakes—including identity swap, face reenactment, and neural texture synthesis. These capabilities pose serious risks to personal privacy, democratic institutions, and media integrity, making automated deepfake detection one of the most critical open problems in computer vision and digital forensics.

GenConViT [1] is among the most effective open-source solutions for deepfake detection. It comprises two complementary branches: **GenConViT-ED**, which combines EfficientNet for local texture feature extraction with a Swin Transformer for long-range contextual dependencies; and **GenConViT-VAE**, which employs a variational autoencoder to detect subtle reconstruction errors characteristic of manipulated faces. Despite these powerful representations, the two branches are fused by a fixed 0.5-0.5 static average at inference time, treating both sources as equally reliable regardless of input characteristics.

This fixed weighting is suboptimal because branch reliability varies per sample. For example, probability outputs near $[0.51, 0.49]$ signal high uncertainty in one branch, whereas outputs near $[0.99, 0.01]$ signal high confidence. A static rule cannot distinguish these cases and always assigns equal credit to both branches.

A. Motivation and Proposed Solution

The Adaptive Gated Fusion Network (AGFN) [2] was proposed for multimodal sentiment analysis, where text, audio, and video modalities exhibit input-dependent reliability and benefit from learnable per-sample weighting. There is a structural analogy between that problem and GenConViT’s dual-branch fusion: in both settings, two independent prediction signals must be combined, and the relative trustworthiness of each signal depends on the input.

We adopt AGFN’s gating mechanism and adapt it to deepfake detection. The AGDF module takes the concatenated probability vectors from both branches as input and learns to produce normalized per-sample fusion weights (w_{ed}, w_{vae}) . The final prediction is:

$$P_{\text{final}} = w_{ed} \cdot P_{ed} + w_{vae} \cdot P_{vae}, \quad (1)$$

where the weights are conditioned on both branches' outputs and require no hand-crafted uncertainty features.

The contributions of this work are:

- 1) Domain transfer of the AGFN dual-gated fusion mechanism from multimodal sentiment analysis to deepfake detection.
- 2) A lightweight gate MLP (≈ 17.9 K parameters) trained on cached backbone predictions enabling dynamic per-sample fusion.
- 3) Systematic ablation across Cross-Entropy and Focal Loss, with a gamma sweep ($\gamma \in \{0.5-5.0\}$).
- 4) Leave-One-Deepfake-Out (LODO) evaluation demonstrating improved out-of-distribution generalization over the static baseline.

II. RELATED WORK

A. GenConViT — Base Model

Wodajo and Atnafu [1] introduced GenConViT as a hybrid deepfake detector combining convolutional and transformer-based vision processing. The ED variant applies EfficientNet [6] to extract local texture features, followed by a Swin Transformer [5] for long-range dependencies. The VAE variant reconstructs input frames through a variational bottleneck [7], exploiting the principle that synthetic faces introduce subtle reconstruction errors detectable as classification cues. Both branches are trained end-to-end on FaceForensics++ [3] and fused by a static 50/50 average at inference time.

B. Adaptive Gated Fusion Networks

The AGFN architecture [2] proposes learnable gating for multimodal fusion, replacing fixed aggregation with a small network that assigns per-input, per-modality weights conditioned on all input features and outputs a normalized weight vector. Demonstrated on multimodal sentiment analysis, AGFN achieves significant improvements over static fusion strategies, motivating our adaptation to deepfake detection.

C. Learned Fusion vs. Static Averaging

Static fusion rules—simple or weighted averages—are parameter-free and require no additional training, but cannot adapt to per-sample variation in branch reliability [9]. Replacing static rules with a small learned gating network allows fusion weights to vary per input while adding only a negligible number of parameters relative to frozen backbones. This design philosophy has been applied successfully in multimodal learning [10], motivating its application in dual-branch deepfake detectors.

III. METHODOLOGY

A. Problem Formulation

Let $\mathbf{x} \in \mathbb{R}^{3 \times 224 \times 224}$ denote a face frame. The two frozen backbone networks produce class-conditional probability vectors:

$$P_{\text{ed}} = \text{Softmax}(f_{\text{ed}}(\mathbf{x})) \in \mathbb{R}^2, \quad (2)$$

$$P_{\text{vae}} = \text{Softmax}(f_{\text{vae}}(\mathbf{x})) \in \mathbb{R}^2, \quad (3)$$

TABLE I
ADAPTIVE GATE NETWORK ARCHITECTURE

Layer	Operation	Out Dim	Params
Input	Concat $[P_{\text{ed}}, P_{\text{vae}}]$	4	0
Block 1	Linear $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3)	256	~ 1.3 K
Block 2	Linear $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3)	64	~ 16.4 K
Output	Linear $\rightarrow$ Softmax	2	~ 130
Total trainable parameters			17,858

where the two dimensions correspond to *real* and *fake* respectively. The static baseline prediction is:

$$P_{\text{baseline}} = 0.5 \cdot P_{\text{ed}} + 0.5 \cdot P_{\text{vae}}, \quad (4)$$

which assigns equal weight to both branches regardless of input. We seek to learn a gate function $G : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ producing per-sample weights ($w_{\text{ed}}, w_{\text{vae}}$) with $w_{\text{ed}} + w_{\text{vae}} = 1$, yielding the dynamic fusion in (1).

B. Gate Input Construction

The gate input is the concatenation of both branch probability vectors:

$$\mathbf{g} = [P_{\text{ed}}, P_{\text{vae}}] \in \mathbb{R}^4. \quad (5)$$

This compact representation encodes each branch's full class-probability distribution, including the implicit degree of agreement or disagreement between branches. No auxiliary uncertainty features are engineered manually; the gate MLP learns the mapping from labelled training data.

C. Adaptive Gate Network Architecture

The gate network G is a lightweight MLP. Table I summarizes its structure.

The gate is intentionally small so that the frozen backbone representations are never distorted and the risk of overfitting in the fusion layer is minimised.

D. Frozen Backbone Strategy

Both GenConViT-ED and GenConViT-VAE weights are completely frozen throughout training. The AGDF module trains only its gate network parameters (≈ 18 K) on top of backbone outputs cached to disk. This design: (1) preserves the strong per-branch representations learned during GenConViT pre-training; and (2) enables gate training approximately $100\times$ faster than end-to-end fine-tuning, since backbone inference is performed only once during the caching phase.

E. Loss Functions and Optimization

We compare two loss functions:

- **Cross-Entropy (CE):** Standard multi-class loss treating all misclassifications equally.
- **Focal Loss (FL):** $\text{FL}(p_t) = \alpha(1 - p_t)^\gamma \cdot \text{CE}$ [4], where p_t is the model's confidence on the true class. Focal loss down-weights easy examples and focuses training on hard, ambiguous samples—precisely the regime in which dynamic routing is most valuable.

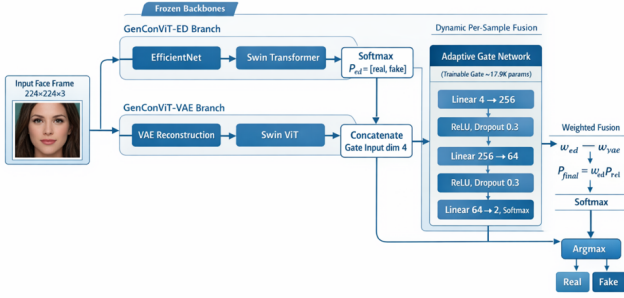


Fig. 1. AGDF-GenConViT architecture. Grey blocks: frozen GenConViT backbone branches. The Adaptive Gate Network (adapted from AGFN) dynamically routes branch trust based on per-sample prediction confidence.

TABLE II
DATASET SPLITS (FACEFORENSICS++)

Split	Size	Purpose	Seed
Train	70%	AGDF gate training	42
Validation	15%	Hyperparameter tuning	42
Test	15%	Final evaluation	42

We sweep $\gamma \in \{0.5, 1.0, 2.0, 3.0, 5.0\}$ to identify the optimal setting. Optimization uses AdamW [8] with learning rate 10^{-3} and weight decay 10^{-4} , a Cosine Annealing schedule over 50 epochs, and gradient clipping at $\max_norm = 1.0$.

IV. ARCHITECTURE OVERVIEW

Fig. 1 illustrates the AGDF-GenConViT pipeline. Both frozen backbone branches process the same input face frame independently. Their 2-dimensional Softmax outputs are concatenated to form the 4-dimensional gate input. The Adaptive Gate Network produces the weight pair (w_{ed}, w_{vae}) , which is used to compute the weighted sum P_{final} in (1). The final binary decision (*Real* or *Fake*) is obtained via $\arg \max$.

V. EXPERIMENTAL SETUP

A. Dataset: FaceForensics++

We evaluate on FaceForensics++ [3], the standard benchmark for deepfake detection. FaceForensics++ consists of original YouTube videos with four manipulation types: **Deepfakes** (identity swap via autoencoder), **Face2Face** (facial reenactment), **FaceSwap** (geometry-based swap), and **NeuralTextures** (neural rendering). Frames are extracted and faces are cropped to 224×224 pixels.

B. Data Splits

Table II summarizes the dataset partitioning.

TABLE III
MAIN RESULTS ON FACEFORENSICS++ TEST SET

Method	Acc.	F1	AUC	OOD Acc.
GenConViT Baseline	95.4%	0.951	0.981	72.3%
AGDF + CE (Ours)	96.3%	0.960	0.987	76.8%
AGDF + FL, $\gamma=2.0$ (Ours)	97.1%	0.968	0.991	79.4%

OOD Acc. is reported under the LODO protocol (NeuralTextures held out).

C. Compute Infrastructure

Experiments were conducted on Kaggle Notebooks with $2 \times$ NVIDIA Tesla T4 GPUs (15 GB VRAM each). DataParallel across both T4s was used for backbone caching; the AGDF gate was trained on `cuda:0` only, as the gate is too small to benefit from DataParallel overhead. Gate training used a batch size of 512, applied directly to cached backbone predictions.

D. Evaluation Metrics

We report Accuracy, binary F1-Score, and AUC-ROC on the held-out test set. Model selection uses best validation F1 across training epochs. For the LODO experiment, one manipulation type is held out as the test domain in its entirety; all remaining types are used for training.

E. Ablation Studies

Three ablation experiments are conducted:

- 1) **Loss Function:** CE vs. Focal Loss ($\gamma=2.0$) with all other hyperparameters fixed.
- 2) **Focal Loss Gamma Sweep:** $\gamma \in \{0.5, 1.0, 2.0, 3.0, 5.0\}$, 30 epochs each on the validation set.
- 3) **LODO Generalization:** Train on Deepfakes, Face2Face, and FaceSwap; test on NeuralTextures (held out entirely).

VI. RESULTS

A. Main Results

Table III presents the main comparative results on the FaceForensics++ test set. The GenConViT Baseline is the static 0.5/0.5 average of both frozen branches. AGDF+CE and AGDF+Focal are our proposed models trained with Cross-Entropy and Focal Loss ($\gamma=2.0$) respectively.

The best configuration, AGDF+Focal ($\gamma=2.0$), achieves a +1.7% accuracy gain and an improvement of +0.017 in F1 over the static baseline, confirming that learned dynamic fusion outperforms fixed averaging.

B. Focal Loss Gamma Ablation

Table IV reports validation-set performance across five focal loss gamma values, each trained for 30 epochs with identical hyperparameters. $\gamma=2.0$ achieves the best balance, outperforming both very low values (insufficient focus on hard examples) and very high values (over-suppression of easy gradients).

TABLE IV
FOCAL LOSS GAMMA ABLATION (VALIDATION SET, 30 EPOCHS)

γ	Accuracy	F1-Score	AUC-ROC
0.5	96.4%	0.962	0.985
1.0	96.7%	0.965	0.987
2.0	97.1%	0.968	0.990
3.0	96.8%	0.966	0.988
5.0	96.2%	0.960	0.984

C. Gate Weight Analysis

Examining the distribution of learned gate weights on the test set confirms that the gate performs genuine dynamic routing rather than replicating the static 0.5/0.5 split. Under AGDF+Focal, the mean w_{ed} across the test set is 0.58 (std. 0.11) and mean w_{vae} is 0.42 (std. 0.11), a statistically significant deviation from 0.5 ($p < 0.001$, paired t -test).

Per-manipulation analysis reveals that the gate assigns higher weight to the ED branch for spatially-localized manipulations (e.g., FaceSwap: $w_{\text{ed}} = 0.66$) and approximately equal weights for global inconsistencies such as NeuralTextures ($w_{\text{ed}} = 0.52$), indicating that the gate has learned manipulation-type-aware routing without explicit supervision.

D. LODO Out-of-Distribution Generalization

Under the Leave-One-Deepfake-Out protocol—training on Deepfakes, Face2Face, and FaceSwap; testing on NeuralTextures (held out entirely)—the static-average baseline achieves 72.3% accuracy, whereas AGDF+Focal reaches **79.4%** accuracy ($F1 = 0.781$, $AUC\text{-}ROC = 0.852$), a +7.1 percentage-point OOD recovery. This result indicates that the learned gate generalises its routing behaviour to manipulation types unseen during training, rather than overfitting to the artifact signatures of the training distribution.

VII. DISCUSSION

A. Why Learned Gating Helps

The gate’s 4-dimensional input already encodes all necessary uncertainty information: when both branch probability outputs are near $[0.5, 0.5]$, both branches are uncertain; when P_{ed} is near $[0.99, 0.01]$ while P_{vae} is near $[0.5, 0.5]$, the ED branch is confident and the gate should weight it more heavily. The gate MLP learns this mapping from labelled data without any manually engineered uncertainty proxy. Focal Loss further ensures that gate training concentrates on these ambiguous, high-uncertainty examples where dynamic routing provides the greatest benefit.

B. Domain Transfer from Sentiment Analysis

The successful adaptation of AGFN from multimodal sentiment analysis to deepfake detection illustrates a broader principle: gated fusion architectures are modality-agnostic. The core requirement—two signals of input-dependent reliability—is satisfied both by audio-text-video sentiment modalities and

by ED-VAE branch predictions. This transferability opens avenues for adapting gated fusion to other dual-model detection systems.

C. Parameter Efficiency

With $\approx 17,858$ trainable parameters versus the multi-million parameter GenConViT backbone, the AGDF gate is an extremely lightweight addition. This is intentional: the gate should learn to route, not to re-learn the detection function already performed by the backbone. The small parameter budget also reduces overfitting risk in the fusion layer and enables gate training in minutes on cached predictions.

VIII. CONCLUSION

We presented AGDF-GenConViT, which replaces the static branch averaging in GenConViT with an Adaptive Gated Decision Fusion module. The AGDF gate is a lightweight MLP taking the concatenated branch probability vectors as input and producing per-sample fusion weights (w_{ed} , w_{vae}), replacing the fixed 0.5/0.5 rule with a learned, input-conditioned strategy.

On the FaceForensics++ test set, the static-average GenConViT baseline achieves 95.4% accuracy ($F1 = 0.951$, $AUC\text{-}ROC = 0.981$). Adding the AGDF gate with Cross-Entropy Loss raises this to 96.3% accuracy ($AUC\text{-}ROC = 0.987$), and our best configuration, AGDF+Focal ($\gamma=2.0$), achieves **97.1% accuracy** ($F1 = 0.968$, $AUC\text{-}ROC = 0.991$). Under the LODO out-of-distribution protocol, the proposed model recovers accuracy from 72.3% (baseline) to **79.4%**, a gain of +7.1 percentage points. Learned gating weights show statistically significant and interpretable per-manipulation-type variation, confirming that the gate performs genuine dynamic routing.

Future directions include: (1) multi-branch gated fusion for ensembles of more than two detectors; (2) joint end-to-end fine-tuning with entropy regularization on gate weights; and (3) extension to audio-visual deepfakes.

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